# Frank Young

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## Personal Summary

- Over 8 years of experience in heterogeneous platform (CPU/GPU/DSP/NPU) operator acceleration library, including DNN, BLAS, FFT, RAND. Deeply involved in 0-1 AI software stack development.
- Three years of team ( 5 people) leadership experience.
- Strong learning ability, pursuit of excellence, and a passion for technology.

## Education

National Taiwan University |Telecom Institute| *Master's Degree* 2014.09—2017.01

[1] **Yang, Hang**, D.-J. Deng, and K.-C. Chen, "On energy saving in ieee 802.11 ax," *IEEE Access*, vol. 6, pp. 47 546–47 556, 2018.

[2] **Yang, Hang**, D.-J. Deng, and K.-C. Chen, "Performance analysis of ieee 802.11 ax ul ofdma-based random access mechanism," in *GLOBECOM 2017-2017 IEEE Global Communications Conference*, IEEE, 2017, pp. 1–6.

Chongqing University |Communication Engineering| *Bachelor's Degree* 2010.09—2014.06

**Top 5%** of the program, outstanding graduate.

## Work Experience

Alibaba Damo |*Senior Software Engineer* 2024.12-Present

- Developed gemm-like operators

Biren Tech |*Senior Software Engineer/Manager* 2021.01—2024.11

- Developed BLAS acceleration libraries from scratch based on GPGPU platform.

Hikvision |*Software Engineer* 2017.07—2020.12

- Developed high-performance convolutional neural network (CNN) library.

## Project Experience

CUTLASS Porting to New Chip |*Programming pattern* 2025.07—Present

- Studied the CUTLASS framework, cute layout algebra, then port to new chip.

BLAS Library Dev |*Development and Maintenance from Scratch* 2023.06—2024.11

- Led the project, responsible for design, construction, development, testing, CICD, documentation, and management.
- Optimized the performance of various large language model GEMM scenarios, benchmarked against A100.
- Designed and implemented kernel selection algorithms.
- Implemented operator registration framework, logging system, and core operator development.

Ultimate Optimization of GEMM |*based on verdi/zebu* 2022.09—2022.11

- Handwritten assembly to achieve ultimate performance GEMM on Vcore, achieved full utilization for specific shapes by observing waveforms through Verdi.
- Implemented simulated FP32 GEMM algorithm based on Tcore, achieved full utilization for specific shapes, and outperformed A100 in performance.

DNN Framework Development |*Self-defined yaml to serialize operator and graph* 2022.03—2023.06

- Developed a deep learning framework.
- Self-defined operator/graph expression based on YAML referencing ONNX.
- Compared precision with mainstream industry solutions and designed custom precision comparison solutions.

## Miscellaneous

- **Technology Stack**: Performance optimization, Acceleration library dev, Deep learning model deployment(CNN, LLM).
- **Languages & Tools**: AI coding copilot, C++, Python, CUDA, CMake, Verdi/SimVision, Git, JIRA, Jenkins.
- **Habits**: Triathon for 5 years.